WS-00: As-Built Physical Airframe & Structural Worksheet

STATIC MECHANICAL BASELINE & AS-BUILT DIMENSIONS (COMPLETE BEFORE INITIAL POWER-ON)

AIRCRAFT: MCO-1D Mark 1	**CARD ID:** WS-00	**DATE:** ____ / ____ / 2026	**LOCATION:** _____
OPERATOR / TECH: _____	**AIRFRAME S/N:** MCO-1D-001	**AMBIENT TEMP:** _____ °C	**BARO / QNH:** _____ hPa

1. Total Airframe Mass Properties & Battery Inventory

Weigh all items on a calibrated digital scale (0.1 g resolution). Record actual values.

Item Description	Location / Component	Nominal Target	As-Built Measurement	Tolerance / Limit
Total Assembled Flying Mass	All 8 wings + 6S & 2S LiPos installed	***< 5,600.0 g*** *(Ref: 5,687 g max)*	[ ] g	Max: 5,687 g Target: 4,800-5,300 g
Dry Airframe Mass	Both LiPo packs removed	*4,972.0 g*	[ ] g	Max: 5,100 g
6S Main Motor LiPo Pack	Tattu 6S 4500mAh 60C XT90-S	*570.0 g*	[ ] g	± 15 g
2S Receiver/Servo LiPo Pack	Gens Ace 2S 3000mAh XT30	*145.0 g*	[ ] g	± 8 g
Carbon Fuselage Spine Assembly	20mm tube + 4 station brackets	*710.0 g*	[ ] g	± 25 g
Flight Control & Avionics Core	Pi 4B + Hat + Pixhawk 4 Mini + Harness	*320.0 g*	[ ] g	± 15 g
Undercarriage & Wheels	Carbon spring struts + Du-Bro wheels	*185.0 g*	[ ] g	± 10 g

2. Individual Wing Panel Weights (S1223 Carbon / LW-PLA Finished Panels)

Bare panels with spar, ribs, covering, tip ballast slug, and root crush ferrule. Must balance port/starboard within 1.5 g.

Wing Station	Port Panel (g)	Starboard Panel (g)	Target Nominal	Port / Stbd Delta	Check (< 1.5 g)
Station 1 (W_1 Fore)	[ ] g	[ ] g	*68.5 g*	[ ] g	[ ] PASS [ ] FAIL
Station 2 (W_2 Mid-Fore)	[ ] g	[ ] g	*66.8 g*	[ ] g	[ ] PASS [ ] FAIL
Station 3 (W_3 Mid-Aft)	[ ] g	[ ] g	*65.5 g*	[ ] g	[ ] PASS [ ] FAIL
Station 4 (W_4 Aft)	[ ] g	[ ] g	*63.2 g*	[ ] g	[ ] PASS [ ] FAIL
TOTALS (All 8 Wings)	**Sum P:** [ ] g	**Sum S:** [ ] g	***528.0 g Total***	**Net: [ ] g**	**Status: [ ] OK**

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GEOMETRY, LONGITUDINAL STAGGER, DRIVETRAIN & BALANCE

3. Wing Semispans & Chord Measurements (Pivot Centerline to Tip)

Wing Station	Port Semi-Span (mm)	Stbd Semi-Span (mm)	Nominal Span	Root Chord (mm)	Tip Chord (mm)
Station 1 (W_1)	[ ] mm	[ ] mm	*772.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 2 (W_2)	[ ] mm	[ ] mm	*749.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 3 (W_3)	[ ] mm	[ ] mm	*735.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 4 (W_4)	[ ] mm		*712.0 mm*	[ ] mm (120)	[ ] mm (120)
Station 1 Total Wingspan	**Tip-to-Tip Span:** [ ] mm		*1,584.0 mm Target*	Fuselage W: [ ] mm	*40.0 mm Target*

4. Cascade Stagger, Drivetrain & Dyneema Rigging Verification

Inspection Parameter	Measurement Method / Tool	Nominal Design Target	As-Built Measurement	Safety Criteria
Station 1 Pivot Axis (x_1)	Steel rule from nose tip	*138.0 mm aft*	[ ] mm	± 1.0 mm
Inter-Station Stagger (s)	Center-to-center pivot distance	*138.0 mm (or 250mm path)*	[ ] mm	± 0.5 mm
Vertical Staircase Step (z)	Vertical drop per station down spine	*41.25 mm drop*	[ ] mm	Matches CAD step
Inter-Station Convergent Slot Gap	Feeler gauge between TE flat & LE tube	*25.0 mm at neutral sweep*	[ ] mm	23.0 to 27.0 mm
Sector Wheel Total Travel	Mechanical stop-to-stop arc (all 8)	*200.0° Total Arc (±100.0°)*	[ ] °	**198.0° to 202.0°**
Flapping Motor Model	12N14P Brushless Motor (8 stations)	*GoolRC 5010-14 360KV (95g)*	[ ] GoolRC [ ] MN5008	**Saves 320g mass**
10:1 Capstan Gear Ratio	Motor turns per full sector arc	*10.00 : 1 Ratio*	[ ] : 1	**Must equal 10:1**
Dyneema SK99 48h Tension	Inline hanging scale (all 8 capstans)	*180.0 N ± 5 N settled*	[ ] N	**Min 160N / Max 200N**
Spar Aluminum Crush Ferrules	Visual check: Ø8x40mm ferrule glued	*Installed all 8 spar roots*	[ ] VERIFIED ALL 8	**Mandatory (PHY-02)**
Motor Mount Thermal Gaskets	Visual: 0.5mm FR4/silicone barrier	*Fitted on all 8 motor mounts*	[ ] VERIFIED ALL 8	**Mandatory (PHY-04)**

5. Center of Gravity (CG) Envelope & Battery Keel Sled Position

Keel Sled Hole Engaged	Measured CG from Nose Tip	Calculated % MAC	Nominal Design Target	Aft Hole & Lockout Roll Pin
Hole # [ ] (Target: 3)	[ ] mm	[ ] % MAC	*26.5% MAC (311.8 mm)*	**[ ] INSTALLED (Locked)**

GATE CERTIFICATION: WS-00: AS-BUILT AIRFRAME MECHANICAL CERTIFICATION	**STATUS: [ ] PASS [ ] CONDITIONAL [ ] FAIL / RE-TEST**
INSPECTOR / LEAD: _____	**SIGNATURE & DATE: _____**
REMARKS / DISCREPANCIES: _____	